POLY PMNA · DIPLOMA CURRICULUM REV2021

Course 4006 | Semester 4 | Project | Shared handbook

Minor Project

A Revision 2021 minor-project guide for programmes assigned to code 4006, with a separate workflow so it is not confused with the 4009 project record.

Course code	**4006**
Revision	**REV2021**
Semester	**Semester 4**
Type	**Project**
Catalogue scope	**11 catalogue programme assignments**

COURSE ORIENTATION

How to use this handbook

This page is a structured learning aid for the shared catalogue record. Study the concept, complete the applied activity, retain the evidence, and compare the result with the latest approved programme instructions. Where the official syllabus differs, the official record takes precedence.

Source and verification note: The course identity, code, semester and subject type are taken from the tracked POLY PMNA REV2021 catalogue, which indexes the official SITTTR scheme. The official course-content reference is SITTTR course 4006; the scheme index is SITTTR REV2021 diploma syllabus. The direct subject-content page was not machine-readable or returned an unavailable-file response during preparation, so module headings and explanatory teaching material on this page are an original study-handbook structure, not claimed verbatim SITTTR wording. Confirm current hours, marks, credits, outcomes and assessment against the latest official programme record before examination use.

Learning outcomes

- Explain the central concepts and vocabulary of the subject using clear technical language.
- Apply a controlled project workflow to a diploma-level problem with scope, milestones and acceptance evidence.
- Create a proposal, plan, design or final evidence pack that a supervisor can review.
- Identify assumptions, limitations and common errors, then improve the work using feedback.

Shared-record context

This code is assigned to 11 catalogue programme assignments in the tracked catalogue. The page therefore focuses on common learning practice and avoids claiming programme-specific technical details that were not verified.

Study rule: Do not treat the author-created examples as official examination questions or official mark distribution.

STUDY MAP

Modules and evidence

Author-created study map; verify official hours and marks			
Module	**Heading**	**Purpose**	**Evidence to retain**
1	Course identity and project boundaries	Establish the code-specific project context and avoid copying the 4009 record or its programme assumptions.	Create a comparison note showing why a 4006 project brief must be approved independently of 4009.
2	Problem selection and approval	Select a useful, feasible problem and present it in a format suitable for supervisor review.	Submit a two-page proposal with a measurable objective, acceptance criteria and resource estimate.
3	Requirements, design and implementation	Move from approved problem to a documented design and a controlled implementation plan.	Build a requirements matrix linking each requirement to a design element and evidence item.
4	Verification and project evidence	Demonstrate that the solution works under stated conditions and document what it cannot do.	Run a peer review using the same evidence a supervisor would need to reproduce the result.
5	Report, demonstration and viva	Present the project concisely and defend individual decisions with evidence rather than unsupported claims.	Complete the final submission checklist and rehearse a three-minute technical explanation.

MODULE 1

Course identity and project boundaries

Purpose-led study | Apply + reflect

Establish the code-specific project context and avoid copying the 4009 record or its programme assumptions.

Core topics−

- Confirm assigned programme and semester
- Project purpose and expected evidence
- Scope boundaries and prohibited overreach
- Academic integrity and safety

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Create a comparison note showing why a 4006 project brief must be approved independently of 4009.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 2

Problem selection and approval

Purpose-led study | Apply + reflect

Select a useful, feasible problem and present it in a format suitable for supervisor review.

Core topics−

- Stakeholder and user need
- Objectives, scope and constraints
- Background study and references

- Approval questions and revision log

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Submit a two-page proposal with a measurable objective, acceptance criteria and resource estimate.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 3

Requirements, design and implementation

Purpose-led study Apply + reflect

Move from approved problem to a documented design and a controlled implementation plan.

Core topics−

- Functional and quality requirements
- Process, block, architecture or data-flow diagrams
- Milestones, responsibilities and version history
- Change requests and decision records

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Build a requirements matrix linking each requirement to a design element and evidence item.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 4

Verification and project evidence

Purpose-led study | Apply + reflect

Demonstrate that the solution works under stated conditions and document what it cannot do.

Core topics−

- Test plan and expected results
- Boundary cases and safety checks
- Defect log and corrective actions
- Limitations, cost and maintainability

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Run a peer review using the same evidence a supervisor would need to reproduce the result.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.

- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 5

Report, demonstration and viva

Purpose-led study

Apply + reflect

Present the project concisely and defend individual decisions with evidence rather than unsupported claims.

Core topics−

- Report chapters and appendices
- Demonstration sequence
- Visual communication and citations
- Individual contribution and likely viva questions

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Complete the final submission checklist and rehearse a three-minute technical explanation.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

ASSESSMENT PREPARATION

Evidence, revision and quality gates

Before submission or examination

1. Confirm the current SITTTR/SBTE title, hours, credits, outcomes and assessment.
2. Define the problem or prompt and state assumptions.
3. Show the method, calculation, algorithm, project record or communication structure.
4. Attach test, source, supervisor, workplace or reflection evidence as applicable.
5. Review clarity, safety, ethics, citations, originality and accessibility.

Revision checklist

- Can I define the main terms without copying?
- Can I apply the method to a new situation?

- Can I explain one common error and its correction?
- Can I show evidence for each claimed outcome?
- Can I state the limitations and the next improvement?

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Evidence record
Subject: 4006 — Minor Project
Task or question:
Assumption or requirement:
Method / design / procedure:
Result or artefact:
Test / source / supervisor evidence:
Reflection and next action:
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SELF-CHECK AND EXAMINATION PRACTICE

Question bank

Recall question

Define two important terms from Minor Project and distinguish them.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Explain question

Explain why the method in this handbook matters for a diploma student.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Apply question

Apply one module method to a realistic technical, project or workplace situation.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Evaluate question

Identify a weak approach, state the evidence needed, and propose a defensible improvement.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Create question

Design a small artefact, plan, test, report section or presentation that demonstrates the subject outcomes.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

REFERENCES AND GLOSSARY

Reference trail

1. Official SITTTR REV2021 diploma syllabus catalogue.
2. Official SITTTR course-content reference for code 4006.
3. POLY PMNA Study Hub, for the current revision index and related lesson pages.

Glossary

Terms used in this handbook

Term	Working meaning
Outcome	A capability the learner should demonstrate, not just a topic read.
Evidence	A record, artefact, result, source, test or review that supports a claim.
Acceptance criterion	A condition used to decide whether a requirement or deliverable is satisfactory.
Reflection	A brief evidence-based account of what worked, what did not and what should change.

Prepared as a POLY PMNA study handbook. Always follow the latest official SITTTR/SBTE instructions for the programme and examination session.